



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATION/ NATIONAL SENIOR CERTIFICATE EXAMINATION

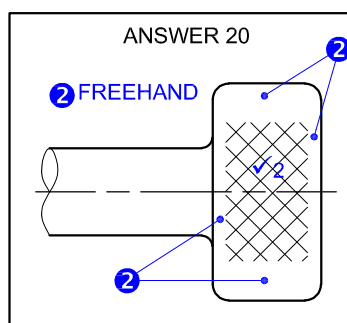
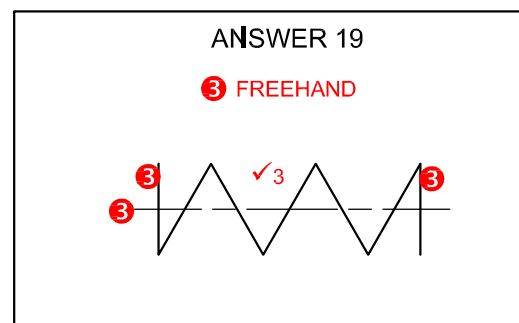
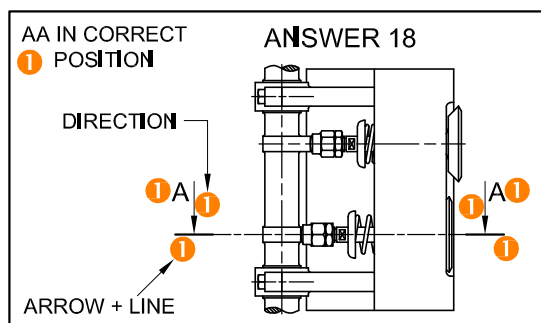
ENGINEERING GRAPHICS AND DESIGN P2

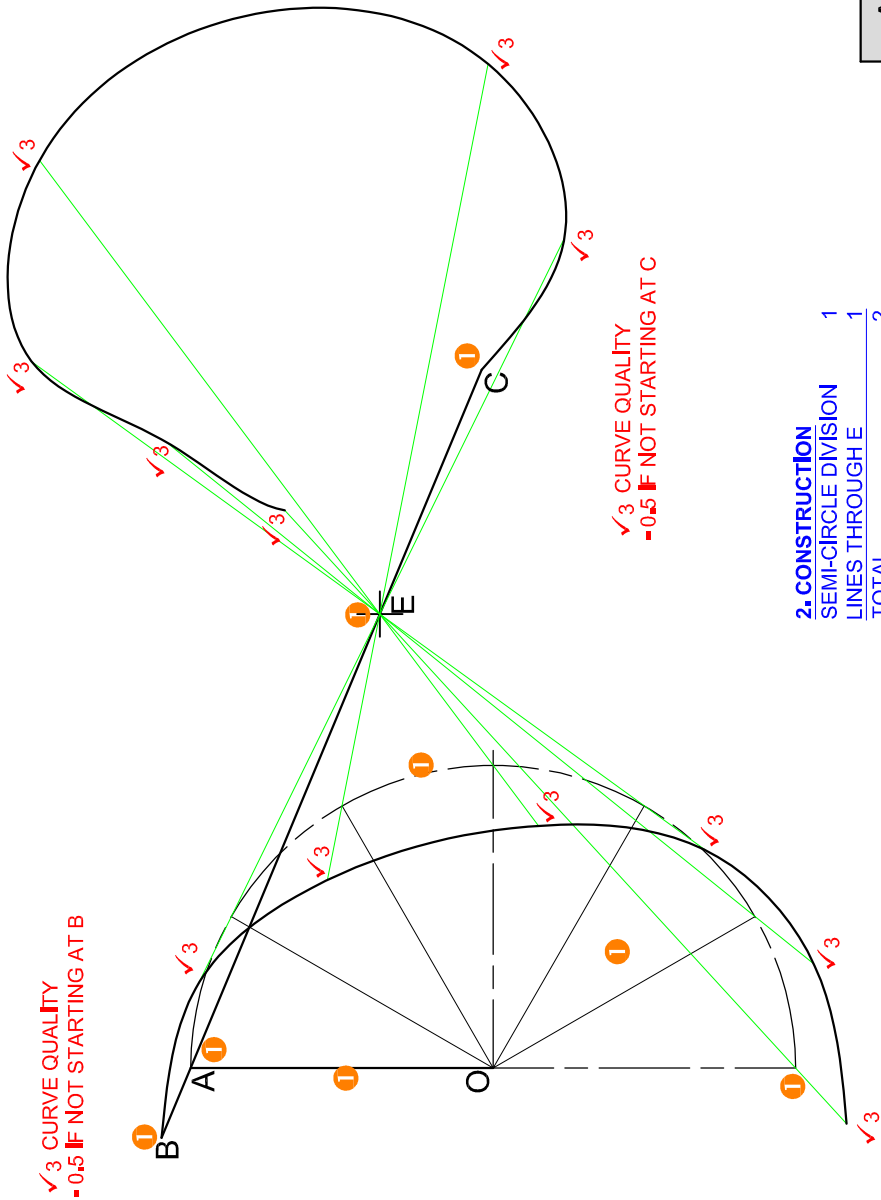
2022

MARKS: 100

These marking guidelines consist of 7 pages.

ANSWERS		
1	MC ENGINEERING WORKS	1
2	GERALD	1
3	4	1
4	02/04/2021	1
5	THIRD ANGLE	1
6	AUTOCAD 2021	1
7	M5	1
8	2	1
9	A: 7 B: R2 C: 15 D: R3 E: AF12	5
10	R15	1
11	FLAT FACE ON A SHAFT	1
12	INTERRUPTED VIEW/CONTINUES	1
13	GRINDING	1
14	39.81	1
15	5 mm	1
16	BOTTOM VIEW	1
17	11	1
18	See below	3 $\frac{1}{2}$
19		3
20		2 $\frac{1}{2}$
TOTAL		30





2. CONSTRUCTION
SEMI-CIRCLE DIVISION 1
LINES THROUGH E 1
TOTAL 2

ASSESSMENT CRITERIA 2.1			
1	GIVEN	4	
2	CONSTRUCTION	2	
3	POINTS + CURVE	14	
TOTAL			20

1. CONSTRUCTION OF GRAPH

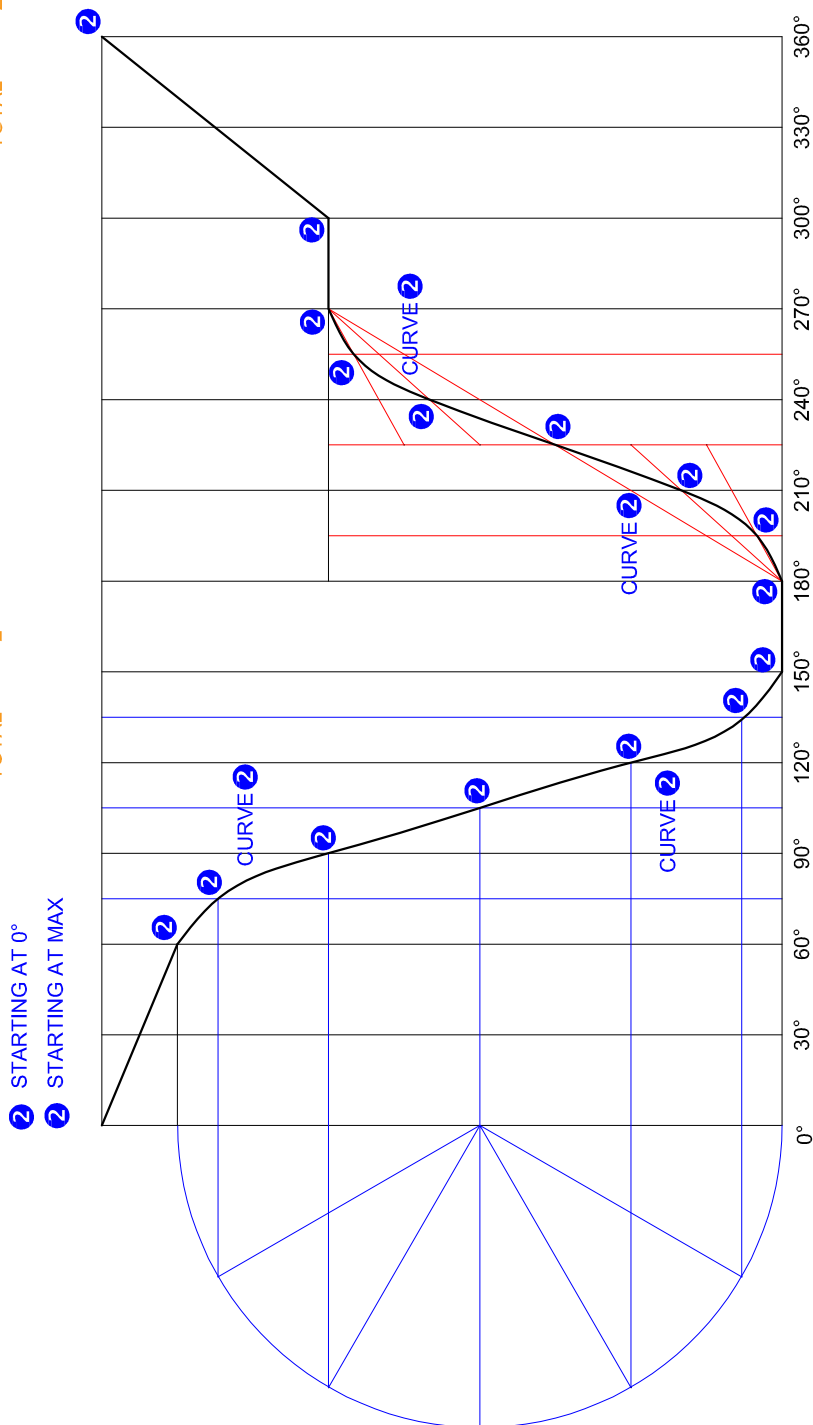
- LENGTH (144) 1
- LINE DIVISION 1 2
- TOTAL 2

1. CONSTRUCTION OF SIMPLE HARMONIC MOTION

- SEMI CIRCLE 0.5
- CIRCLE DIVISION 0.5
- ADDITIONAL 15° DIVISIONS 0.5
- HORIZONTAL PROJECTION 0.5
- TOTAL 2

1. CONSTRUCTION OF UNIFORM ACCELERATION + RETARDATION

- ADDITIONAL 15° DIVISIONS 0.5
- VERTICAL DIVISION 0.5
- 1st HALF 0.5
- 2nd HALF 0.5
- TOTAL 2

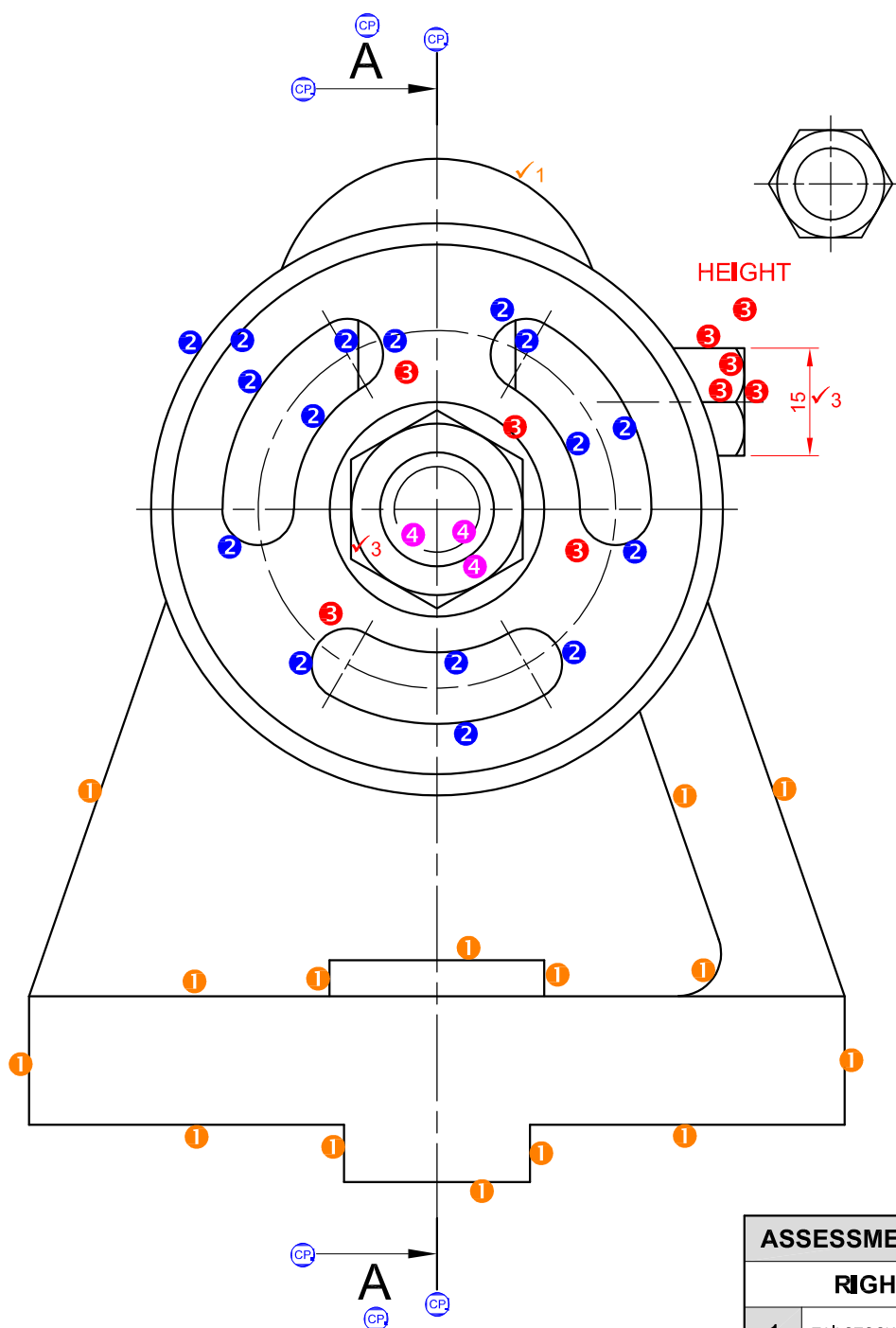


② STARTING AT 0°

② STARTING AT MAX

DISPLACEMENT GRAPH ③
SCALE: 144 mm = 360° ROTATION ③

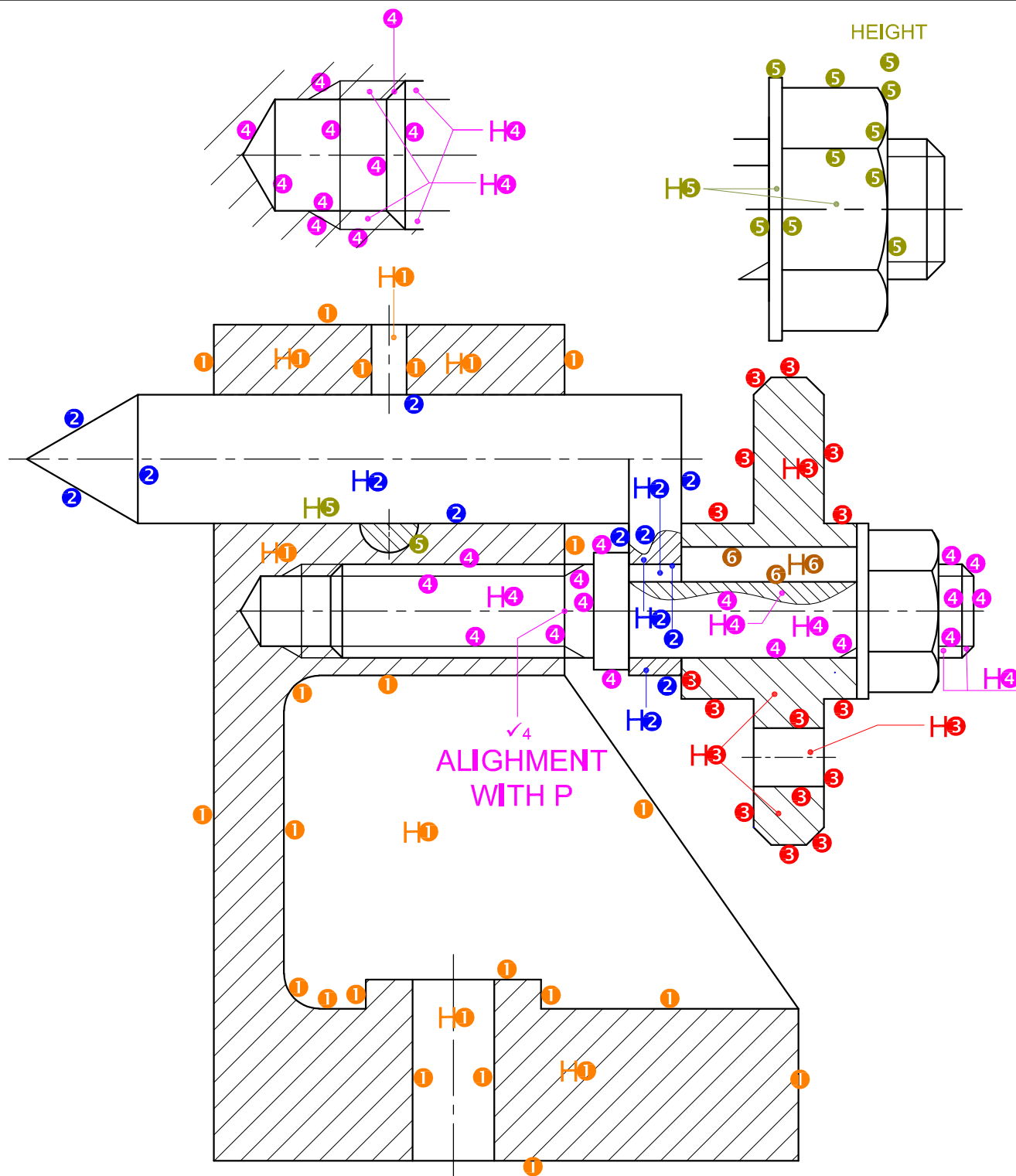
ASSESSMENT CRITERIA 2.2		
1	CONSTRUCTION	6
2	POINTS + CURVE	11
3	LABELS	1
TOTAL		18



3. ASSEMBLY (ALTERNATIVE IF ONLY THE RIGHT VIEW WAS DRAWN)
1 MARK FOR EVERY COMPONENT CORRECTLY ASSEMBLED (7 PARTS - 1) = 6

ASSESSMENT CRITERIA		
RIGHT VIEW		
1	TAILSTOCK	8 $\frac{1}{2}$
2	ADJUSTING WHEEL + DEAD CENTRE	8
3	M16 NUT + WASHER + LOCKING NUT	6 $\frac{1}{2}$
4	ADJUSTING STUD	1 $\frac{1}{2}$
SUBTOTAL		24 $\frac{1}{2}$
GENERAL		
2	CUTTING PLANE	3

PAPER 2 QUESTION 4
GRADE 12
2022
MARKING GUIDELINES

**3. ASSEMBLY**

1 MARK FOR EVERY COMPONENT CORRECTLY ASSEMBLED (8 PARTS $\times$ 1) = 7

1. CENTRE LINE MARK ALLOCATION

SANS COMPLAINT (LINE TYPE)

CENTRE LINES

(1 to 4 = 1 / 5 to 11 = 2 / 12 to 14 = 3)

TOTAL 4

ASSESSMENT CRITERIA		
GENERAL		
1	CENTRE LINES	4
3	ASSEMBLY	7
SUBTOTAL		14

PAPER 2 QUESTION 4

GRADE 12

2022

MARKING GUIDELINES

ASSESSMENT CRITERIA**SECTIONAL FRONT VIEW**

1	TAILSTOCK	14
2	DEAD CENTRE	7
3	ADJUSTING WHEEL	9
4	ADJUSTING STUD	16 $\frac{1}{2}$
5	M16 NUT + WASHER + M10 LOCKING BOLT	6 $\frac{1}{2}$
6	KEY	1 $\frac{1}{2}$
SUBTOTAL		54 $\frac{1}{2}$